Independent

$p_{XY} = p_X p_Y$	$p_{X\|Y} = p_X$
$F_{XY} = F_X F_Y$	$F_{X\|Y} = F_X$
$E(XY) = E(X)E(Y)$	$E(X\|Y) = E(X)$
$D(X+Y) = D(X) + D(Y)$	$Cov(X,Y) = 0$
$H(X+Y) = H(X) + H(Y)$	$I(X,Y) = 0$

$$\begin{tikzcd}
p_X \arrow[r, shift left, "\cdot p_{Y|X}"] \arrow[rr, bend left, "\text{Bayes}"] \arrow[d, "\int_0^x"'] & p_{XY} \arrow[l, shift left, "\int dy"] \arrow[r, "/ p_Y"] & p_{X|Y} \arrow[ll, bend left, "\text{Full Possi}"] \arrow[d, "\partial_x"] \\
F_X \arrow[r, shift left, "\cdot F_{Y|X}"] & F_{XY} \arrow[l, shift left, "F(x,\infty)"] & F_{X|Y}
\end{tikzcd}$$

$$p_{X|Y} = \frac{p_{XY}}{p_Y}$$

$$p_X = \int p_Y p_{X|Y} dy \qquad \text{Full Possi}$$

$$p_{X|Y} = \frac{p_X p_{Y|X}}{p_Y} = \frac{p_X p_{Y|X}}{\int p_X p_{Y|X} dy} \qquad \text{Bayes}$$

e.g. 由 F_{XY} 求 $F_{X|Y}$

$$F_{X|Y} = \int_0^x p_{X|Y} = \int_0^x \frac{p_{XY}}{p_Y}$$
$$= \int_0^x \partial_x \partial_y F_{XY} / \partial_y F_Y$$
$$= \partial_y F_{XY} / \partial_y F_{XY}(\infty, y)$$

e.g. 由 $F_{X|Y}$ 求 F_{XY}

$$F_{XY} = \int_0^x \int_0^y p_{XY} = \int_0^x \int_0^y p_{X|Y} \cdot p_Y$$
$$= \int_0^x \int_0^y \partial_x F_{X|Y} \; p_Y$$
$$= \int_0^y F_{X|Y} \cdot p_Y$$

Exception $E(X) = \int x f(x) dx$

Variance $D(X) = E(X^2) - [E(X)]^2 = E[(X-\mu)^2]$

$D(X+Y) = D(X) + D(Y) + 2Cov(X,Y)$

Correlation $R(X,Y) = E(XY)$

Covariance $C(X,Y) = E(XY) - E(X)E(Y) = E[(X-\mu_X)(Y-\mu_Y)]$ $\qquad \rho_{XY} = \frac{cov}{\sigma_X \sigma_Y}$

$E(X|Y) = \int x p(x|y) dx$ $\qquad$ $E(X|Y)$ 是 Y 的函数

Iteritive Exception Formula $E[E(X|Y)] = E(X)$

Charicristic Function $\Phi(\omega) = \mathcal{F}p(-\omega)$

$$= E(e^{j\omega x}) = \int p(x)\, e^{j\omega x} dx$$

e.g. $X \sim Exp(\lambda)$, $\Phi(\omega) = \dfrac{\lambda}{\lambda - j\omega}$

e.g. $X \sim N(\mu, \sigma^2)$, $\Phi(\omega) = e^{j\omega\mu - \frac{1}{2}\sigma^2\omega^2}$

$$\Phi(\omega) = \int p_X \cdot e^{j\omega x} dx = \frac{1}{\sqrt{2\pi}\,\sigma} \int e^{-\frac{1}{2\sigma^2}[(x-\mu)^2 - 2\sigma^2 j\omega x]} dx$$

$$x^2 - (2\mu + 2\sigma^2 j\omega)x + \mu^2$$

$$[x - (\mu + \sigma^2 j\omega)]^2 - 2\mu\sigma^2 j\omega + \sigma^4\omega^2$$

$$= e^{j\omega\mu - \frac{1}{2}\sigma^2\omega^2} \cdot \frac{1}{\sqrt{2\pi}\,\sigma} \int e^{-\frac{1}{2\sigma^2}[x - (\mu + \sigma^2 j\omega)]^2} dx$$

$X \sim N(\mu, \sigma^2)$ $\quad E(e^{tX}) = e^{t\mu + \frac{1}{2}t^2\sigma^2}$

MGF (Moment Generating Func)

$$\varphi(s) = \mathcal{L}p(-s) = \int p(x)\, e^{sx} dx$$

$$E(X^k) = \frac{1}{j^k}\Phi'(0) = \varphi'(0)$$

PGF (Prob Generating Func)

$$G(z) = \mathcal{Z}p(z^{-1}) = \sum p(k) \cdot z^k$$

$$p(n) = \frac{1}{n!} G^{(n)}(0)$$

$E(X) = G'(1)$ $\quad D(X) = G''(1) + E(X) - E^2(X)$

PDF of $X+Y$ 可先求 $\Phi_Z = \Phi_X \Phi_Y$ 或 $\varphi_Z = \varphi_X \cdot \varphi_Y$

$x(t)$	$X(\omega)$	$X(f)$	$X(s)$
1	$2\pi\delta(\omega)$	$\delta(f)$	
$\delta(t)$	1	1	
$e^{j\omega_0 t}$	$2\pi\delta(\omega-\omega_0)$	$\delta(f-f_0)$	
$\delta(t-t_0)$	$e^{j\omega t_0}$	$e^{-j2\pi f t_0}$	
$e^{-\lambda t}u(t)$	$\frac{1}{j\omega+\lambda}$		$\frac{1}{s+\lambda}$
$te^{-\lambda t}u(t)$	$(\frac{1}{j\omega+\lambda})^2$		$(\frac{1}{s+\lambda})^2$
$e^{-\lambda\lvert t\rvert}$	$\frac{2\lambda}{\omega^2+\lambda^2}$		
$\frac{2\lambda}{t^2+\lambda^2}$	$2\pi e^{-\lambda\lvert\omega\rvert}$	$2\pi e^{-\lambda 2\pi\lvert f\rvert}$	
$\cos\omega_0 t$	$\pi[\delta(\omega-\omega_0)+\delta(\omega+\omega_0)]$	$\frac{1}{2}[\delta(f-f_0)+\delta(f+f_0)]$	
$\sin\omega_0 t$	$\frac{\pi}{j}[\delta(\omega-\omega_0)-\delta(\omega+\omega_0)]$	$\frac{1}{2j}[\delta(f-f_0)-\delta(f+f_0)]$	

$X(f)$ is $\omega \to 2\pi f$

$2\pi\delta(\omega) \to \delta(f)$

$\mathcal{L}$ is $\mathcal{F}$ $j\omega \to s$